

Another critical hurdle is the lack of interoperability among existing systems, which will significantly increase the complexity and costs of IoT deployments. Present-day technologies work largely in silos. However, in the future, a full-functional digital ecosystem will require seamless data sharing between machines and other physical systems from different manufacturers.

Moving forward

Although it is perceived that value-added services such as applications and system integrations will be the significant revenue driver, hardware is also likely to generate sizeable revenue through its sheer volume of deployment. This would no doubt impact the demand for electronic components. Individual IoT devices will inevitably require a microcontroller (MCU) to add intelligence to the device, one or more sensors to allow for data collection, one or more chips for connectivity and data transmission, and a memory component. Connected

devices that transmit information across relevant networks rely on innovation from semiconductor players—for instance, highly-integrated microchip designs and very-low-power functions.

A new array of sensors based on microelectromechanical systems (MEMS) technology is rapidly being developed to enable IoT applications beyond motion and image sensing. These would include applications that measure humidity, altitude, food calorie composition and various human health indicators.

With connectivity being a key capability required in IoT applications, multiple communication standards and protocols are being supported by consumer electronics and industrial devices, like low-power and personal area wireless mesh networks, which are primarily suited for sensors. IoT applications usually require a single chip in an acceptable form

factor and low power consumption for battery-operated devices. This requirement is a factor for growth in SoCs with embedded cores, GPUs and integrated wireless connectivity in a single package. Progress in these technologies has allowed the development of such IoT devices as sensors that have compute, storage and network capabilities built into extremely small form factors, yet with low energy requirements.

India should look to build capabilities across areas such as sensors (to adapt to rugged climate/terrains in India) along with network infrastructure, standards and so on. Accelerators and incubators in India are already enabling startups to build innovative IoT solutions. Addressing current talent gaps in terms of cross-functional as well as specialised skillsets is imperative for the growth of the IoT in India. **END** 

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Continued from page 84...

roles at home, work, and in public are conceived. 'Respectable femininity', which is a refraction of hegemonic masculinity, has provided women with the 'right amount of freedom' to participate in the lucrative IT industry so long as their essential identity as wives and mothers was not impacted.

Women's exclusion and discrimination in the Indian FOSS community is rooted in a different set of gendered assumptions and priorities from those in the West, where it is assumed women are not as intellectually capable as men. While Indian women are generally presumed to be as technically capable as men (because much of the work of the Indian FOSS community such as evangelising is social rather than technical and done in public, social settings), many women cannot participate because they do not have as much free time away from

work and family obligations as men do. Further, due to the prevalence of homosociality, many women and men do not necessarily feel comfortable socialising together in the public sphere.

For the most part, the Indian FOSS community is left-leaning, middle class and urban, though there are significant generational, regional, and gendered differences. The older generation (aged 50 +) is almost exclusively upper middle class and upper caste, and its members have held important positions in the government, academia, R&D, NGOs, etc. The younger generation, in its 20s and 30s, is most likely to be students or employed by IT companies, with a few entrepreneurs as well. In general, class, caste and gender inequalities are maintained in the FOSS community, though many members of the community are sincerely committed to making it more inclusive.

FOSS offers Indians the possibility of exerting autonomy in the relationships

between technology and the state. FOSS allows for the autonomy of states in relation to the market and it also, crucially, offers autonomy to citizens in relation to markets and the Indian state. The Indian FOSS community has taken a technology created in a Western context and mobilised it towards what can broadly be called nation building efforts, though to what extent these mobilisations lead to substantive societal change is yet to be seen.

Note: Interested readers can obtain a copy of the entire thesis by writing to Dr Jasmine Folz at jasminefolz@gmail.com.

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